

Case Reference: CASE-2026-INIT
Transaction ID: TXN-INIT-001
Timestamp: 2026-08-16 20:20:00 UTC
Amount: **\$2,450.00 USD**
Action Taken: AUTO_BLOCK

RISK LEVEL: CRITICAL
93.0/100

Status: Immediately terminate transaction, flag account, and notify Cardholder.

1. Multi-Factor Risk Model Diagnostics

Diagnostic Factor	Raw Metric	Factor Weight	Normalized Impact
Supervised Machine Learning (XGBoost)	94.0% Fraud Probability	65%	61.1 / 65.0
Unsupervised Anomaly (Isolation Forest)	87.5 Outlier Index	25%	21.9 / 25.0
Domain Heuristics & Rules	3 Rule(s) Triggered	10%	10.0 / 10.0
CALIBRATED COMPOSITE RISK SCORE	CRITICAL SEVERITY	100%	93.0 / 100.0

2. Explainable AI (SHAP) Forensic Feature Attribution

Primary Behavioral Triggers Identified:

- Elevated latent fraud index score (+4.90)
- Critical latent identity / card-use deviation (V14 = -3.82)
- Critical latent identity / card-use deviation (V12 = -3.99)
- Significant deviation on feature V4 (2.21)
- Critical latent identity / card-use deviation (V10 = -3.54)

Feature Name	Observed Value	SHAP Attribution (Delta)	Risk Direction
Risk_Indicator_Index	23.5291	+4.9014	Elevates Fraud Probability
V14	-3.8229	+3.0472	Elevates Fraud Probability
V12	-3.9934	+1.3381	Elevates Fraud Probability
V4	2.2131	+1.2444	Elevates Fraud Probability
V10	-3.5387	+1.1070	Elevates Fraud Probability

3. Fraud Analyst Investigation Notes & Audit Signature

Investigator: Lead Fraud Analyst #8142

Findings: System automated pipeline validation dossier. Verified all models operational.

Digital Audit Checksum (SHA-256):
e52052c4b782390426ecd558b485cdbb0143efafacf7d35e3fad007ab6449b4b

Verified Fraud Risk Engine
COMPLIANCE CERTIFIED